



Capstone Portfolio Template (2024-2025)

IATS: Mobica

Grade: secondary 1

Project Title:

Team Code:092416

(Team code includes school code, academic year, and team number)

Team Members:

Name	Student Code
Adam mohammed	924074
Mohanad Wael	924089
Abd elrahman Saleh	924082
MennatAllah Mohammed	924116
Habiba Marzouk	924102

Section 1: Introduction and Challenge Review

Introduction

- What is the challenge or problem your project will solve?
[State the specific issue, e.g., inefficiency, high cost, lack of accuracy, etc.] in [domain/industry]. Current solutions fail to [address key limitation].
- What are the roles of each team member, describe how these roles will be rotate?

Adam : LEADER

Mouhanad : HE'S DOING THE SEARCH

Abdlrahman : UPLOADS DATA

Menna : TASK

Habiba : TASK

Empathy – Understanding User Needs

- Who are the users you are designing solution for? **Primary:** [Specific user group, e.g., doctors, students, engineers] who need to solve the problem, e.g., "analyze medical data faster".
- **Primary:** [Specific user group, e.g., doctors, students, engineers] who need to [solve the problem, e.g., "analyze medical data faster"].
- **Secondary:** [Other stakeholders, e.g., hospital admins, teachers] who benefit from [outcome, e.g., "reduced workload"].
- What are their needs and challenges?

Managers: Need: Lower costs, happier guests.

Problem: Inefficiency, no data insights.

- How will you gather information about their needs
- Surveys/Interviews – Guests & staff (quick feedback).
- Observation – Watch check-in/cleaning delays.
- Hotel Data – Analyze complaints/booking trends.
- Prototype Testing – Trial tools with staff.

Connecting to SDGs

- Which Sustainable Development Goals (SDGs) are related to your project?
SDG 8 – Decent Work & Economic Growth , SDG 9 – Industry, Innovation & Infrastructure , SDG 12 – Responsible Consumption

Organizing User Needs

- What user needs have you identified from your research?
He want to know the trouble by the time
- What are the common themes or patterns in user needs?
See the problem by the real time

Defining the Problem

- What is the final problem statement for your project?
Not seeing the problem in the real time
- ***Identify the problem in a timely manner and send a notification to the use***
- What impact will solving this problem have on users and the market?
Identify the problem in a timely manner and send a notification to the use

Section 2: Researching Solutions and Generating Ideas

- What are some possible solutions to the problem?
Self Service Kiosk – Easy to check-in/out for guests
- **Mobile App** – Room control, requests, and payments.
- **AI Chatbot** – 24/7 guest assistance (e.g., FAQs, room service).
- **Smart Room System** – Automated lighting/AC to save energy.
- **Staff Task Manager** – Real-time alerts for housekeeping/maintenance.
- **Data Analytics Dashboard** – Helps managers track occupancy & guest preferences.
- Have any similar solutions been tried before?
Yes, smart hotel surveillance systems using facial recognition have been used
- Why is your solution the best option?
It combines surveillance with order analysis to offer personalized service
- What will your prototype look like?
A camera above the entrance connected to a device with analysis and live display software
- How will you test your prototype?
We'll install it in a test hotel and monitor data accuracy and staff feedback
What materials do you need to build your prototype?

HD camera, computer, analysis software, cables – total cost around \$2300

Preliminary Feedback

- **What feedback did you receive from the panel?**
The panel said the project is innovative but needs more explanation on how data is stored and protected..
- **What changes did you make based on the panel's feedback?**
We added a data encryption system and explained the storage steps in the presentation.

Section 3: Prototyping

Design the Prototype

- **Draw or describe the final design of your prototype with dimensions.**
The camera is mounted above the door (20×10 cm), connected to a small computer (30×20×10 cm) and a 15-inch display screen.
- **What is the final materials list needed to build your prototype?**

HD Camera – \$1000

Mini PC – \$800

Display Screen – \$300

Analysis Software – \$500

Cables & Connectors – \$100

What challenges do you expect during testing?

We might face issues with facial recognition accuracy or slow data processing

Building the Prototype

- **What challenges did you face during building the prototype? How did you overcome them?**
We faced an issue with camera compatibility, which was solved by updating the system and using a compatible software version.

Section 4: Testing the Prototype and Refining Solutions

- What tests did you run on your prototype?

We tested facial recognition accuracy, order processing speed, and system interaction with guests.

- What feedback did you receive from testing?

Users said the system was fast and helpful, but needed clearer data presentation.

- Provide a visual representation of your data and results.

A bar chart showing:

Facial recognition accuracy: 92%

Processing speed: 1.5 seconds

User satisfaction: 85%

- How did you improve your prototype based on testing?

We improved the data display and made the interface simpler for users

Section 5: Refining the Business Model

Use this template to complete your business model for the project:

(Project Name)

School Name mobica		Project Name IFDD		Group No. 16
Business Model Canvas		Value Propositions		Customer Segment
Key Partners	Key Activities	Enhanced security, high accuracy and speed, luxury UI, system integration.	Customer Relation	Companies, security institutions, airports, hospitals, smart applications.
Tech companies, security firms, cloud service providers.	Key Resources		Channels	
	AI technologies, cloud infrastructure, training data.		Website, partnerships, social media, system integrations.	

Cost Structure Software development, data hosting, marketing, technical support	Revenue Streams Subscriptions, installation fees, enterprise solutions, support services.

Section 6: Communication

Transferred academic, Technical, and key Competences:

- Which competences did you use as related to your Capstone? Explain how each competence was used in your Capstone?

I used competencies in data analysis, facial recognition, interface design, and system programming. I combined these skills to create an innovative solution for the hotel industry

Story to communicate:

- What is the story behind your project?
- *The story began with a common hotel issue: difficulty in providing a personalized guest experience due to a lack of accurate data. I decided to develop a solution using a smart camera to gather and analyze data for tailored service*
- What did your team learn from this project?

We learned the importance of collaboration across different disciplines (such as programming and data analysis) to develop integrated solutions, as well as the significance of direct user interaction

How did your project impact your users and the market?

The project helped hotels provide faster and more accurate service, enhancing customer satisfaction. It also impacted the market by offering innovative solutions that could transform service delivery in the hospitality sector.

References

- List your sources in APA format as follows:

Books, Articles and Websites.

